

# A plot tells a thousand words

UNDERSTANDING DATA VISUALIZATION



**Richie Cotton**

Data Evangelist at DataCamp

# What you'll learn

- How do you choose an appropriate plot?
- How do you interpret common types of plots?
- What are best practices for drawing plots?

# Three ways of getting insights

## Calculating summary statistics

mean, median, standard deviation

## Running models

linear and logistic regression

## Drawing plots

scatter, bar, histogram

# The Datasaurus Dozen

| away_x | away_y | bullseye_x | bullseye_y | ... | x_shape_x | x_shape_y |
|--------|--------|------------|------------|-----|-----------|-----------|
| 32.33  | 61.41  | 51.20      | 83.34      | ... | 38.34     | 92.47     |
| 53.42  | 26.19  | 58.97      | 85.50      | ... | 35.75     | 94.12     |
| 63.92  | 30.83  | 51.87      | 85.83      | ... | 32.77     | 88.52     |
| 70.29  | 82.53  | 48.18      | 85.05      | ... | 33.73     | 88.62     |
| 34.12  | 45.73  | 41.68      | 84.02      | ... | 37.24     | 83.72     |
| 67.67  | 37.11  | 37.89      | 82.57      | ... | 36.03     | 82.04     |

<sup>1</sup> Matejka, J., & Fitzmaurice, G. (2017) <https://www.autodeskresearch.com/publications/samestats>

# Mean of x for each dataset

| dataset    | mean(x) |
|------------|---------|
| away       | 54.27   |
| bullseye   | 54.27   |
| circle     | 54.27   |
| dino       | 54.26   |
| dots       | 54.26   |
| h_lines    | 54.26   |
| high_lines | 54.27   |

| dataset    | mean(x) |
|------------|---------|
| slant_down | 54.27   |
| slant_up   | 54.27   |
| star       | 54.27   |
| v_lines    | 54.27   |
| wide_lines | 54.27   |
| x_shape    | 54.26   |

# Mean of x and y for each dataset

| dataset    | mean(x) | mean(y) |
|------------|---------|---------|
| away       | 54.27   | 47.83   |
| bullseye   | 54.27   | 47.83   |
| circle     | 54.27   | 47.84   |
| dino       | 54.26   | 47.83   |
| dots       | 54.26   | 47.84   |
| h_lines    | 54.26   | 47.83   |
| high_lines | 54.27   | 47.84   |

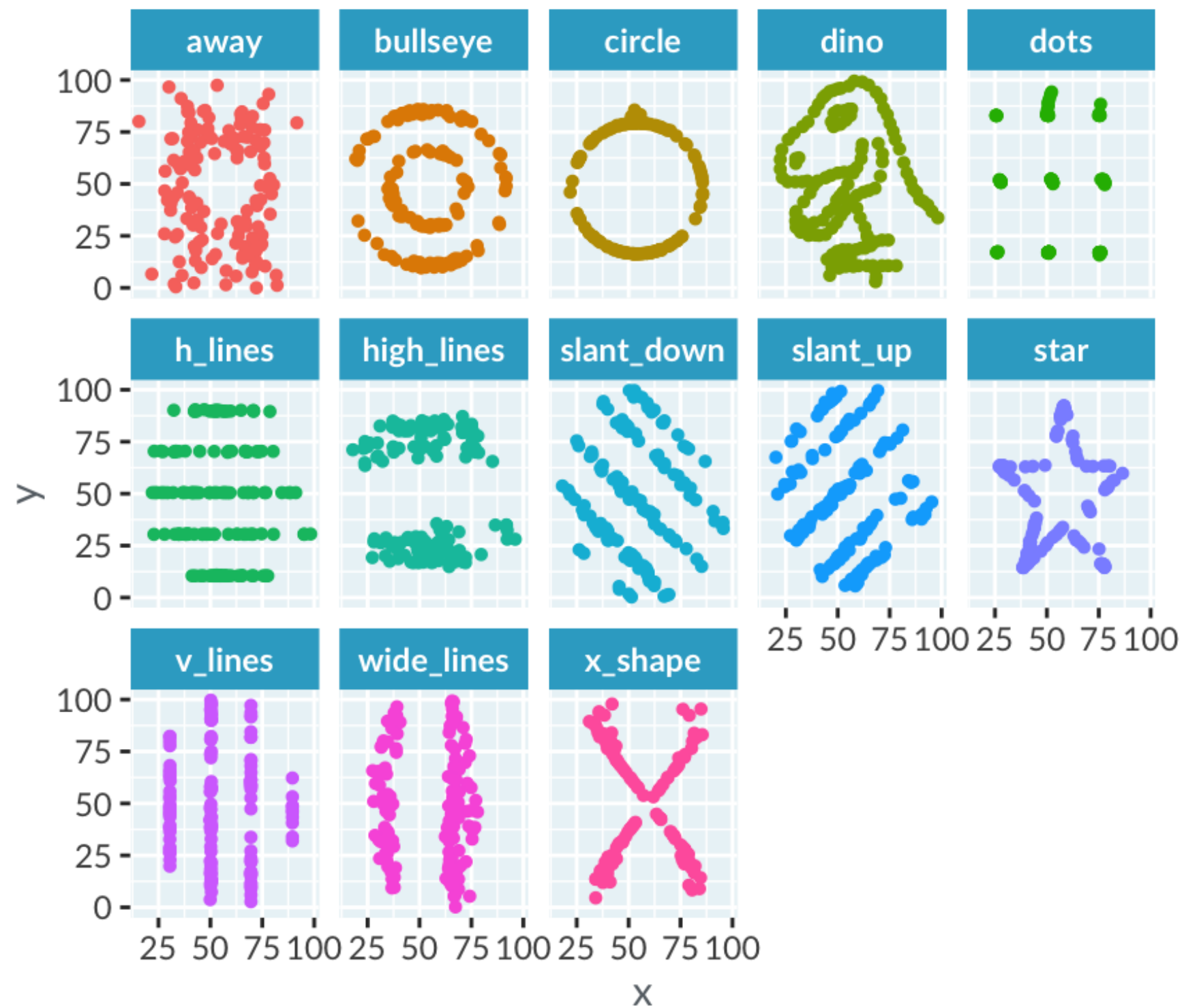
| dataset    | mean(x) | mean(y) |
|------------|---------|---------|
| slant_down | 54.27   | 47.84   |
| slant_up   | 54.27   | 47.83   |
| star       | 54.27   | 47.84   |
| v_lines    | 54.27   | 47.84   |
| wide_lines | 54.27   | 47.83   |
| x_shape    | 54.26   | 47.84   |

# Standard deviations for each dataset

| dataset    | std_dev(x) | std_dev(y) |
|------------|------------|------------|
| away       | 16.77      | 26.94      |
| bullseye   | 16.77      | 26.94      |
| circle     | 16.76      | 26.93      |
| dino       | 16.77      | 26.94      |
| dots       | 16.77      | 26.93      |
| h_lines    | 16.77      | 26.94      |
| high_lines | 16.77      | 26.94      |

| dataset    | std_dev(x) | std_dev(y) |
|------------|------------|------------|
| slant_down | 16.77      | 26.94      |
| slant_up   | 16.77      | 26.94      |
| star       | 16.77      | 26.93      |
| v_lines    | 16.77      | 26.94      |
| wide_lines | 16.77      | 26.94      |
| x_shape    | 16.77      | 26.93      |







# Continuous and categorical variables

## Continuous: usually numbers

- heights, temperatures, revenues

# Continuous and categorical variables

## Continuous: usually numbers

- heights, temperatures, revenues

## Categorical: usually text

- eye colors, countries, industry

# Continuous and categorical variables

## Continuous: usually numbers

- heights, temperatures, revenues

## Categorical: usually text

- eye colors, countries, industry

## Can be either

- age is continuous, but age group is categorical
- time is continuous, month of year is categorical

# Let's practice!

UNDERSTANDING DATA VISUALIZATION

# Histograms

UNDERSTANDING DATA VISUALIZATION



**Richie Cotton**

Data Evangelist at DataCamp

# When should you use a histogram?

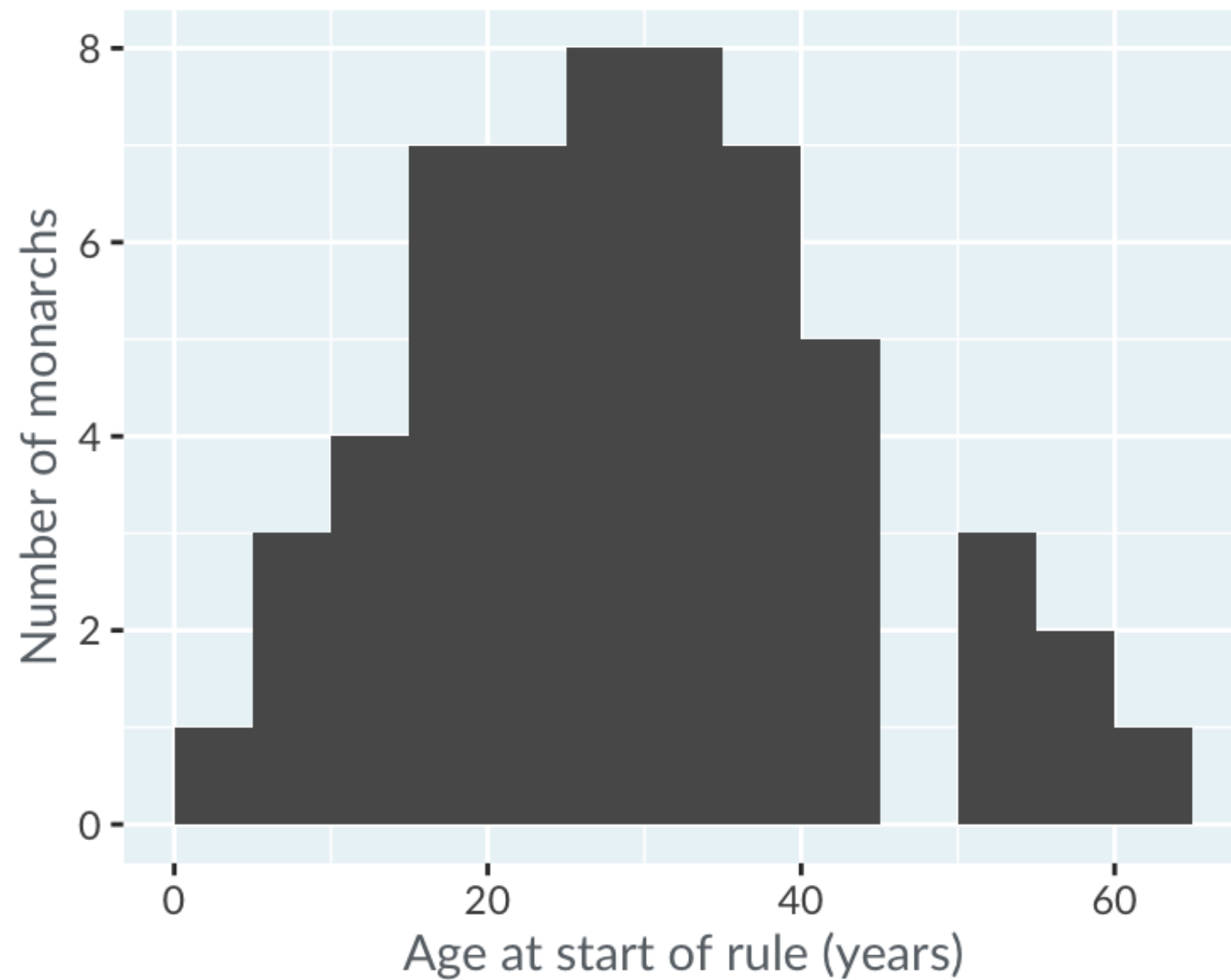
1. If you have a single continuous variable.
2. You want to ask questions about the shape of its distribution.

# Kings and Queens of England & Britain

| official_name | house   | birth_date | start_of_rule | age_at_start_of_rule |
|---------------|---------|------------|---------------|----------------------|
| Charles III   | Windsor | 1948-11-14 | 2022-09-08    | 73.86575             |
| Elizabeth II  | Windsor | 1926-04-21 | 1952-02-06    | 25.79603             |
| George VI     | Windsor | 1895-12-14 | 1936-12-11    | 40.99110             |
| Edward VIII   | Windsor | 1894-06-23 | 1936-01-20    | 41.57426             |
| ...           | ...     | ...        | ...           | ...                  |
| Eadred        | Wessex  | 0923-07-01 | 0946-05-26    | 22.90212             |
| Edmund I      | Wessex  | 0921-07-01 | 0939-10-27    | 18.32170             |
| Aethelstan    | Wessex  | 0894-07-01 | 0924-07-01    | 29.99863             |

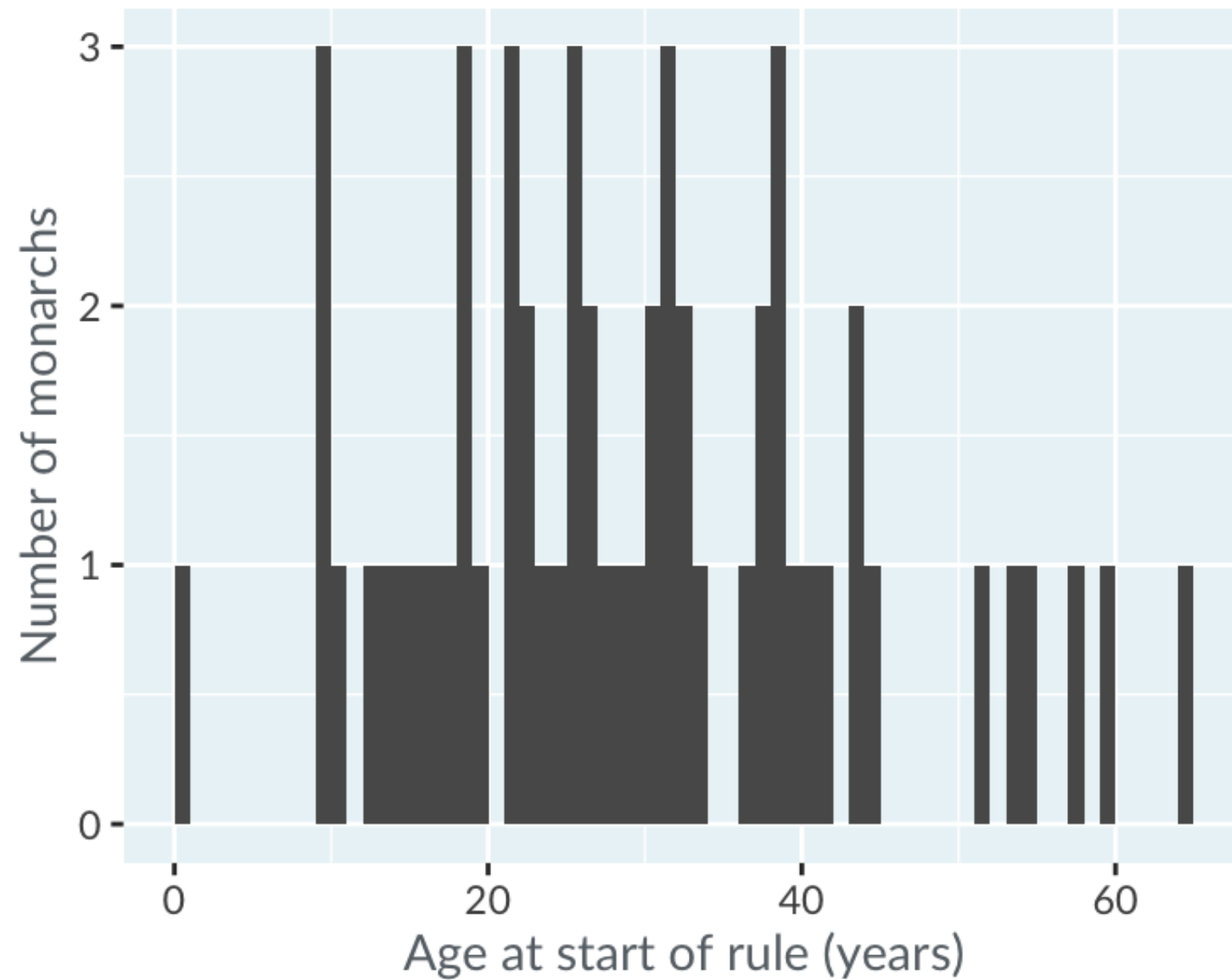


# Histogram of age at start of rule

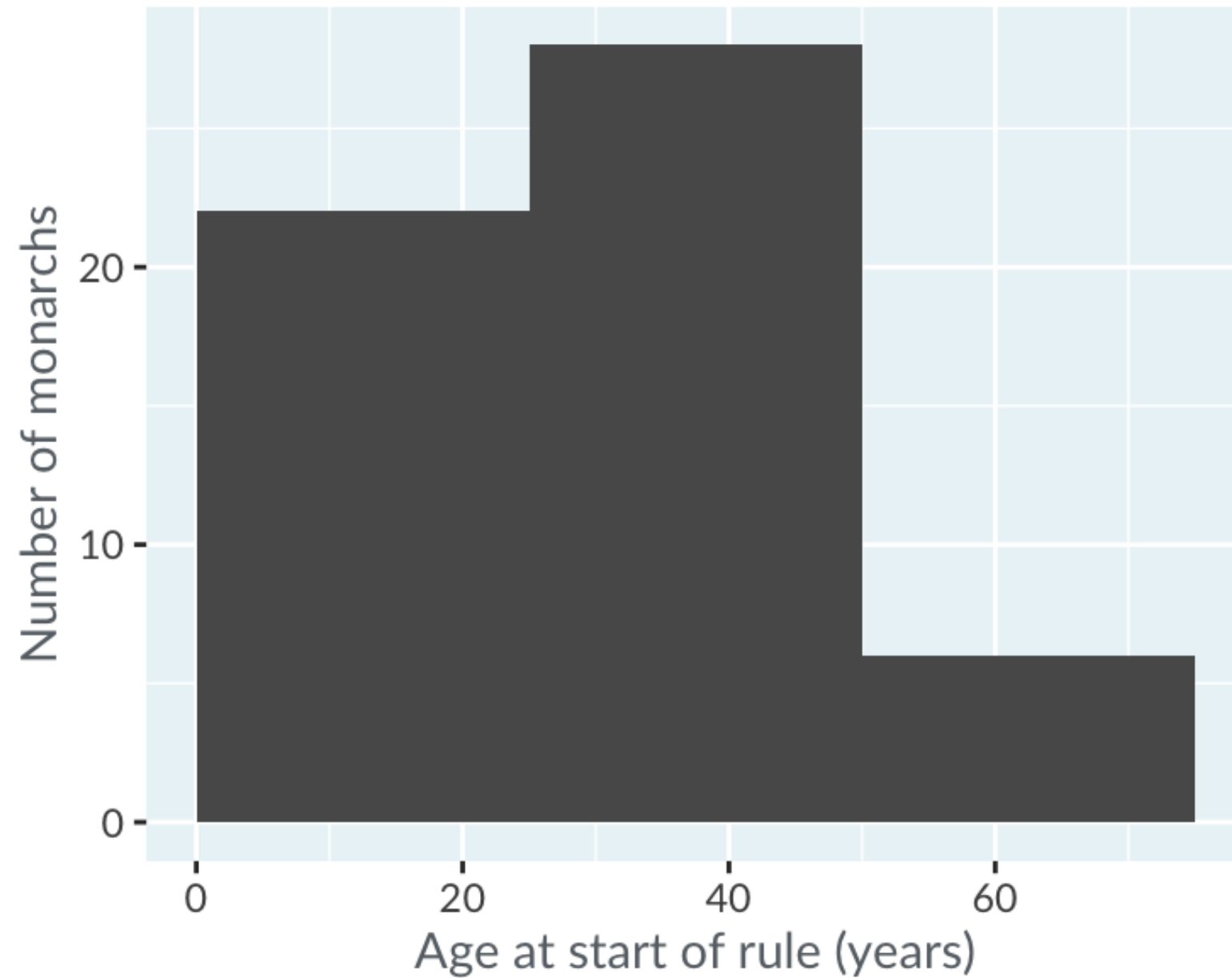




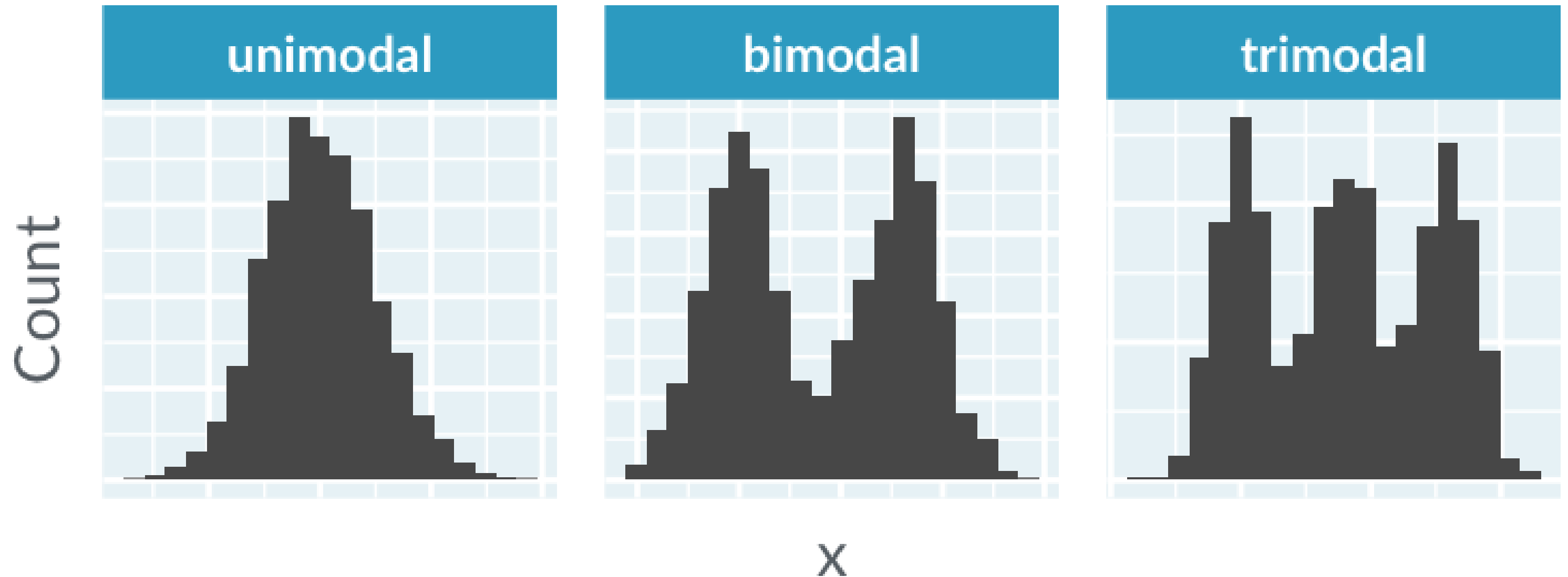
# Choosing binwidth: 1 year



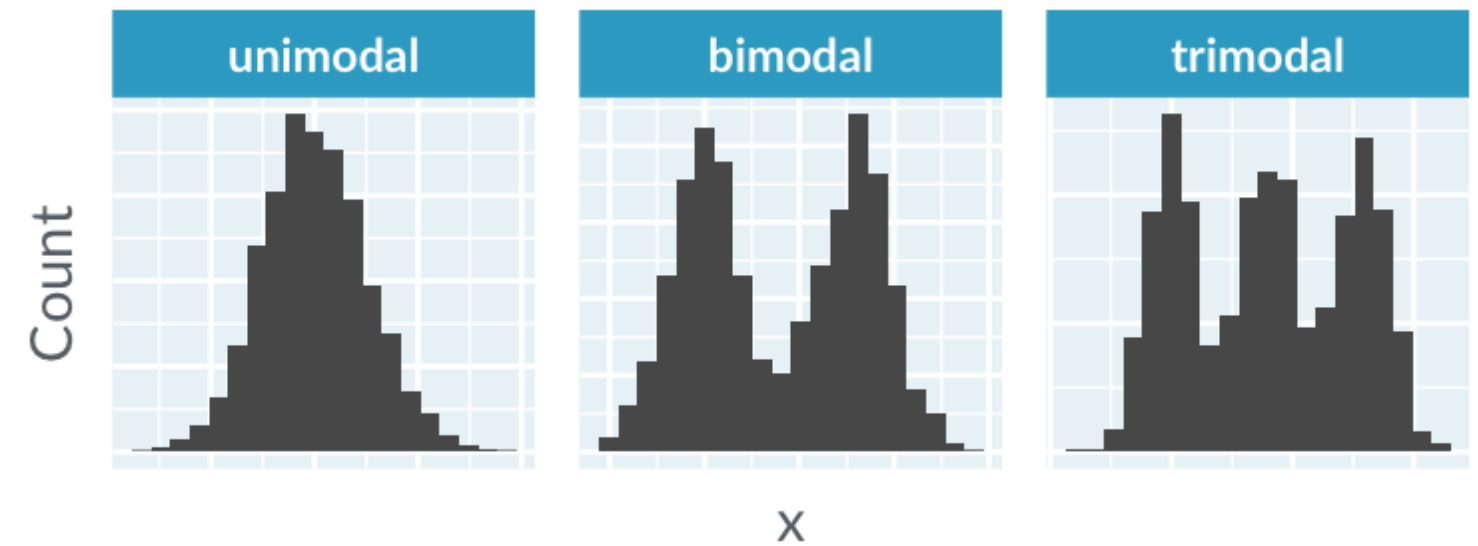
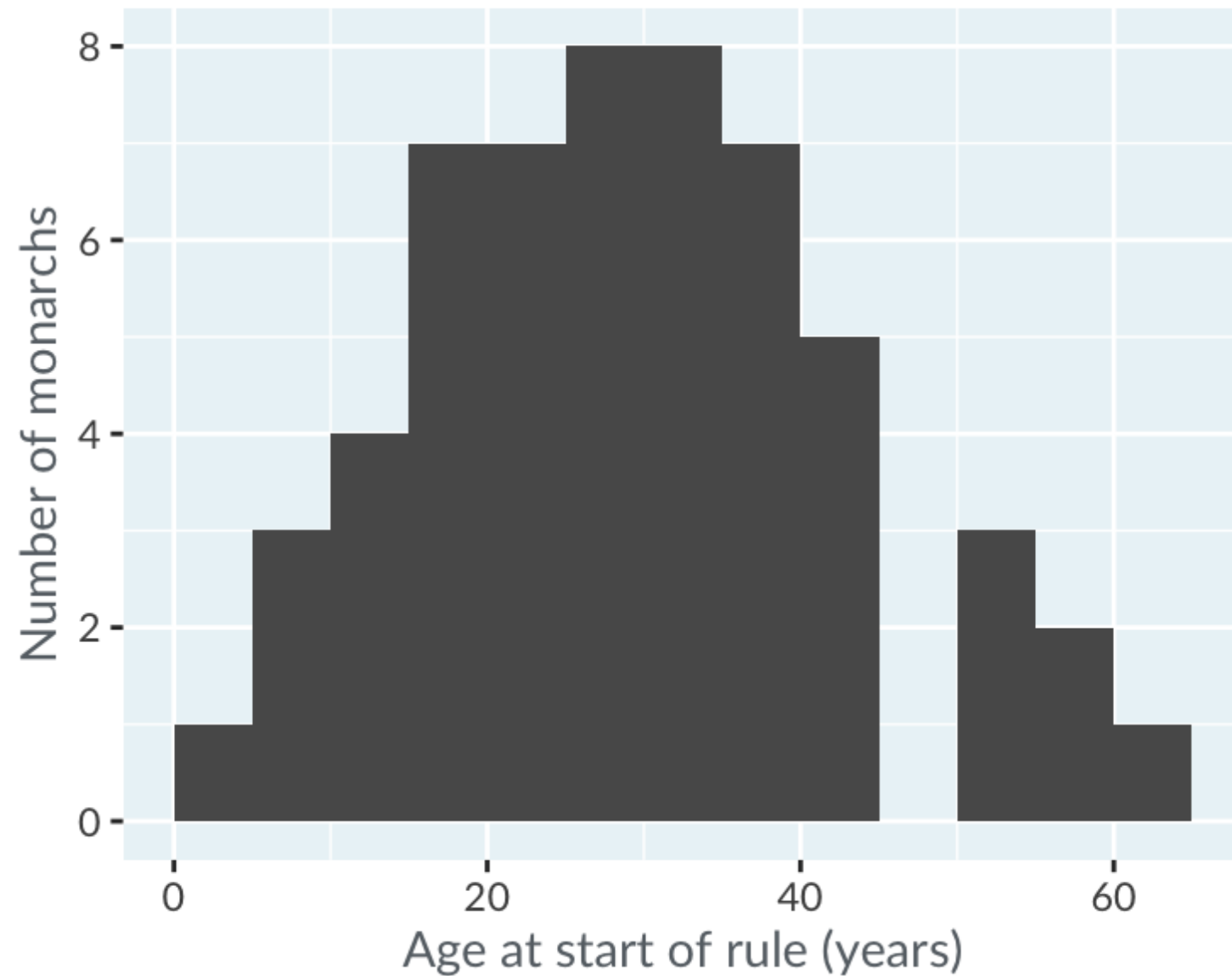
# Choosing binwidth: 25 years



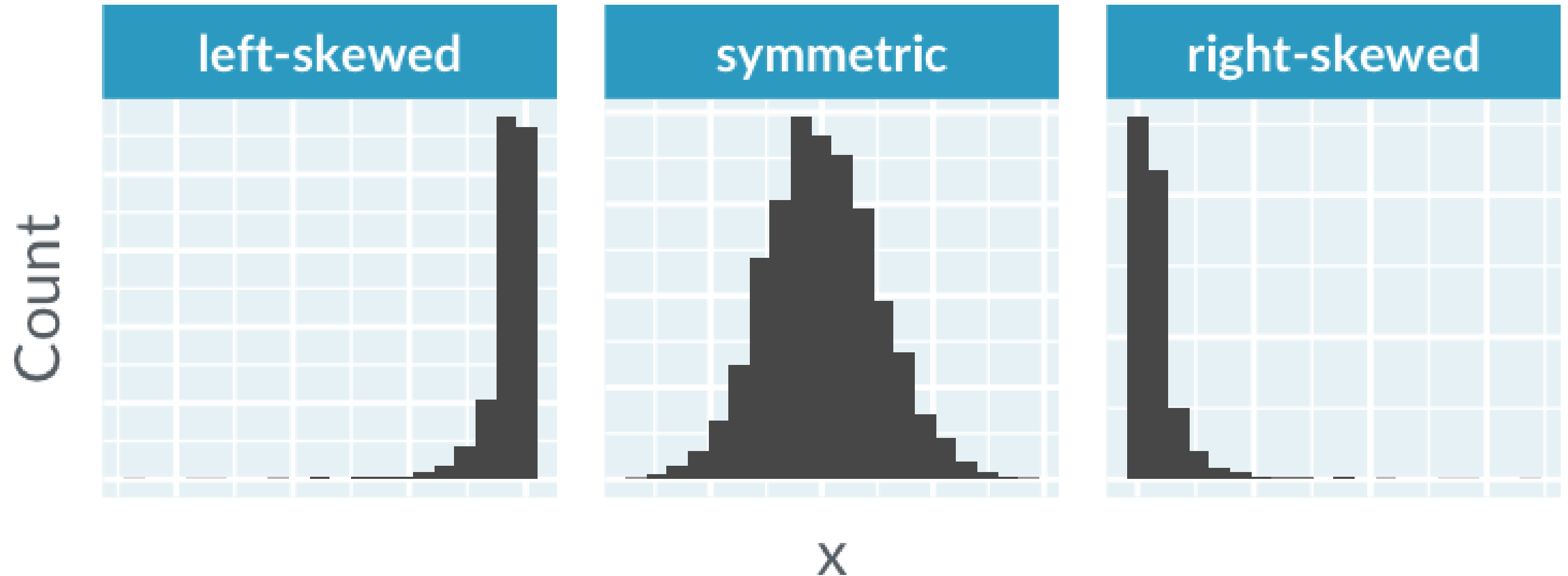
# Modality: how many peaks?



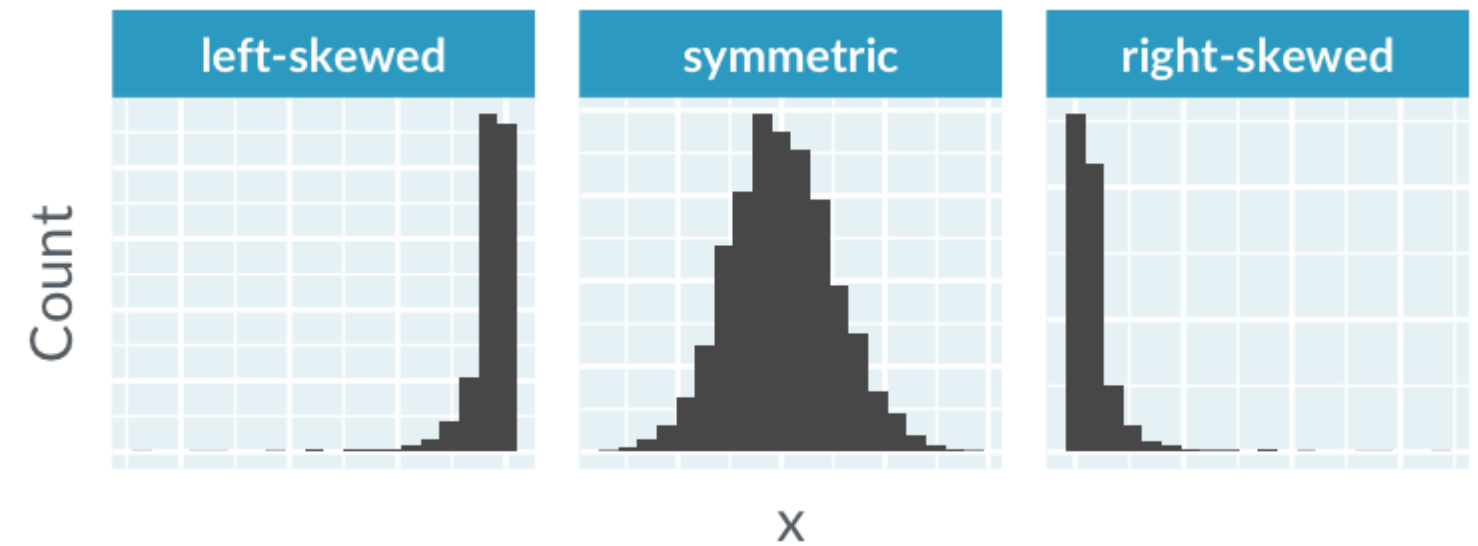
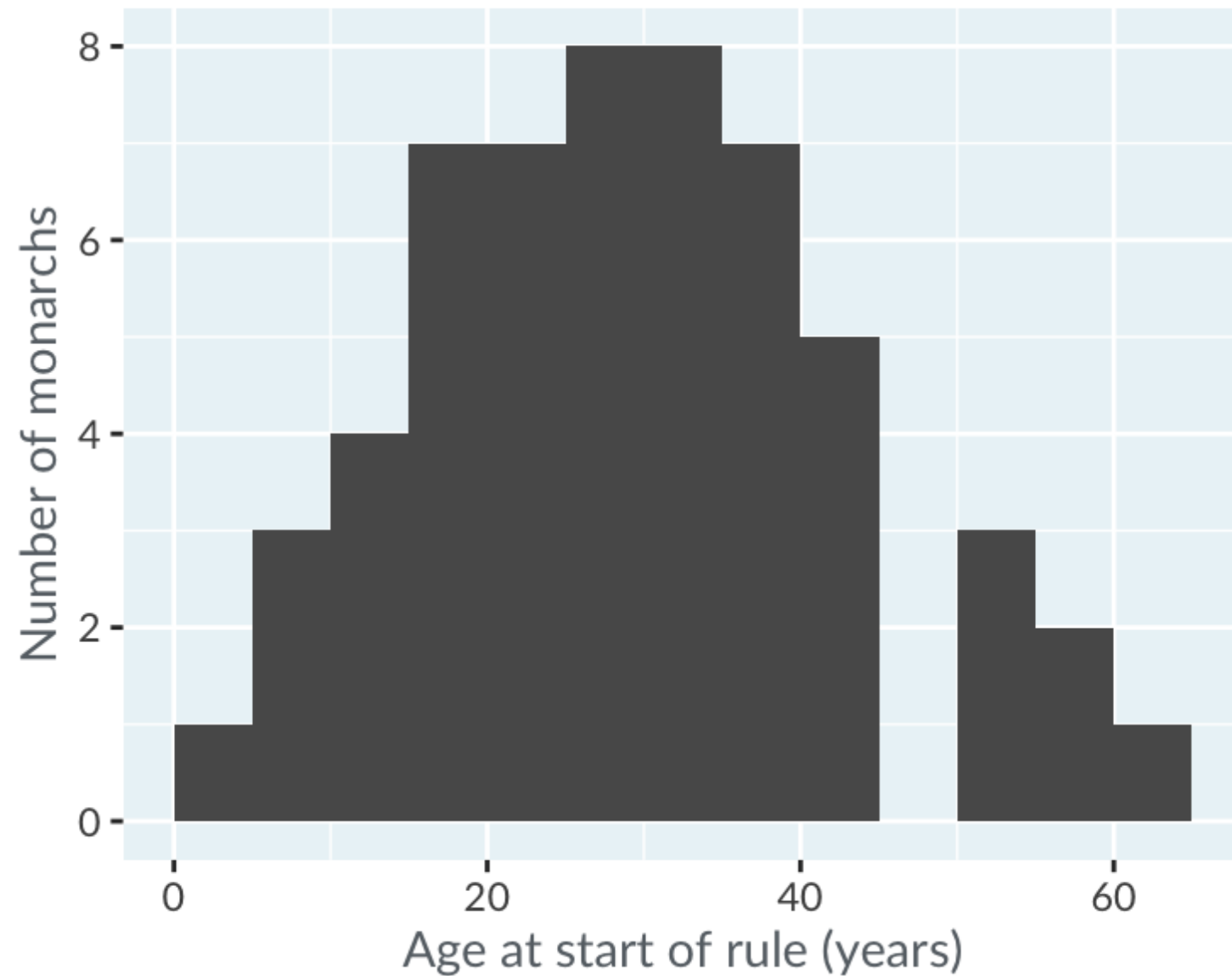
# Modality: how many peaks?



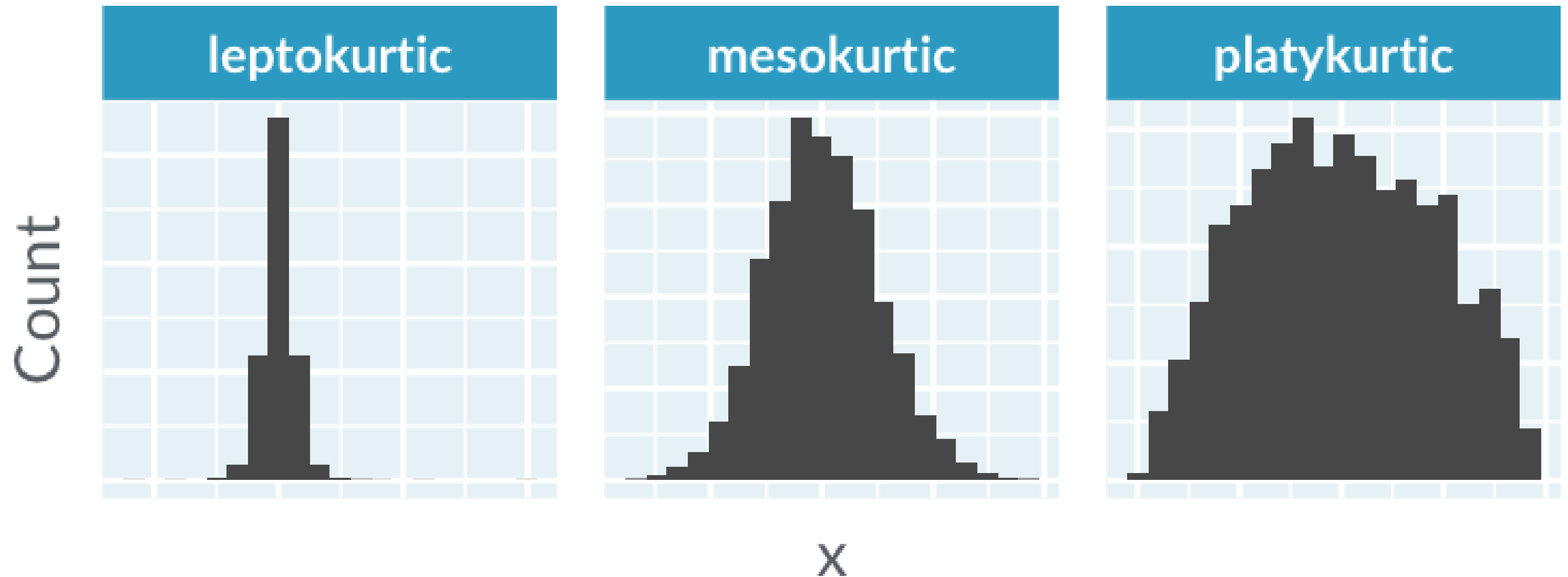
# Skewness: is it symmetric?



# Skewness: is it symmetric?

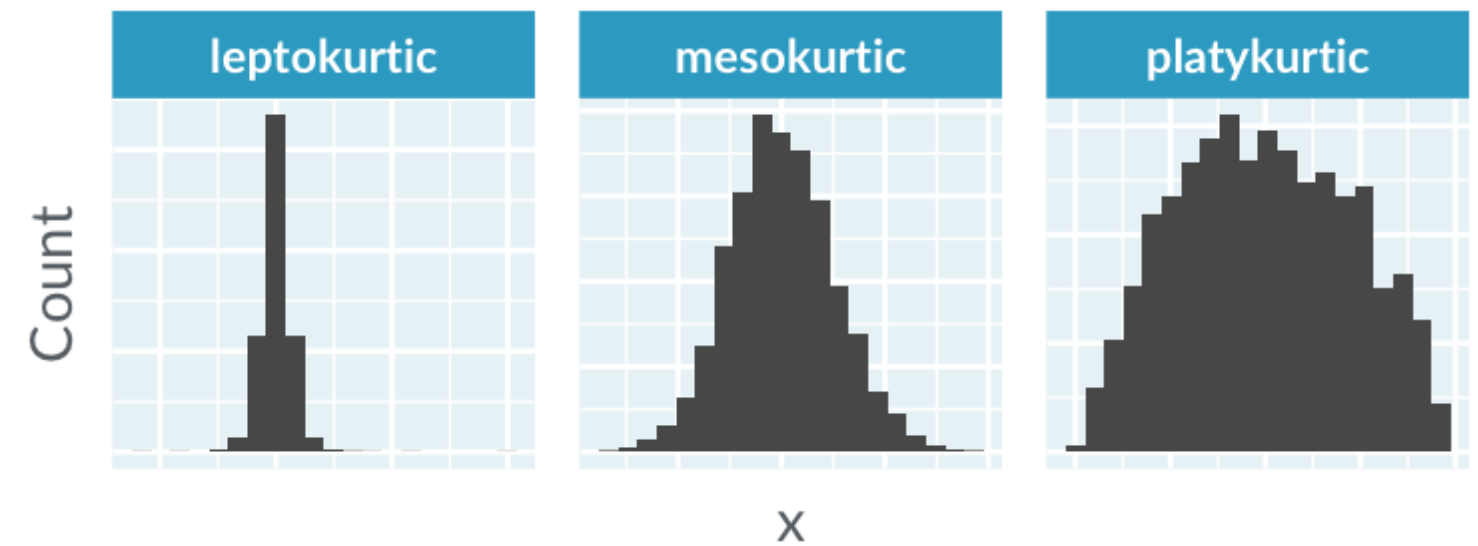
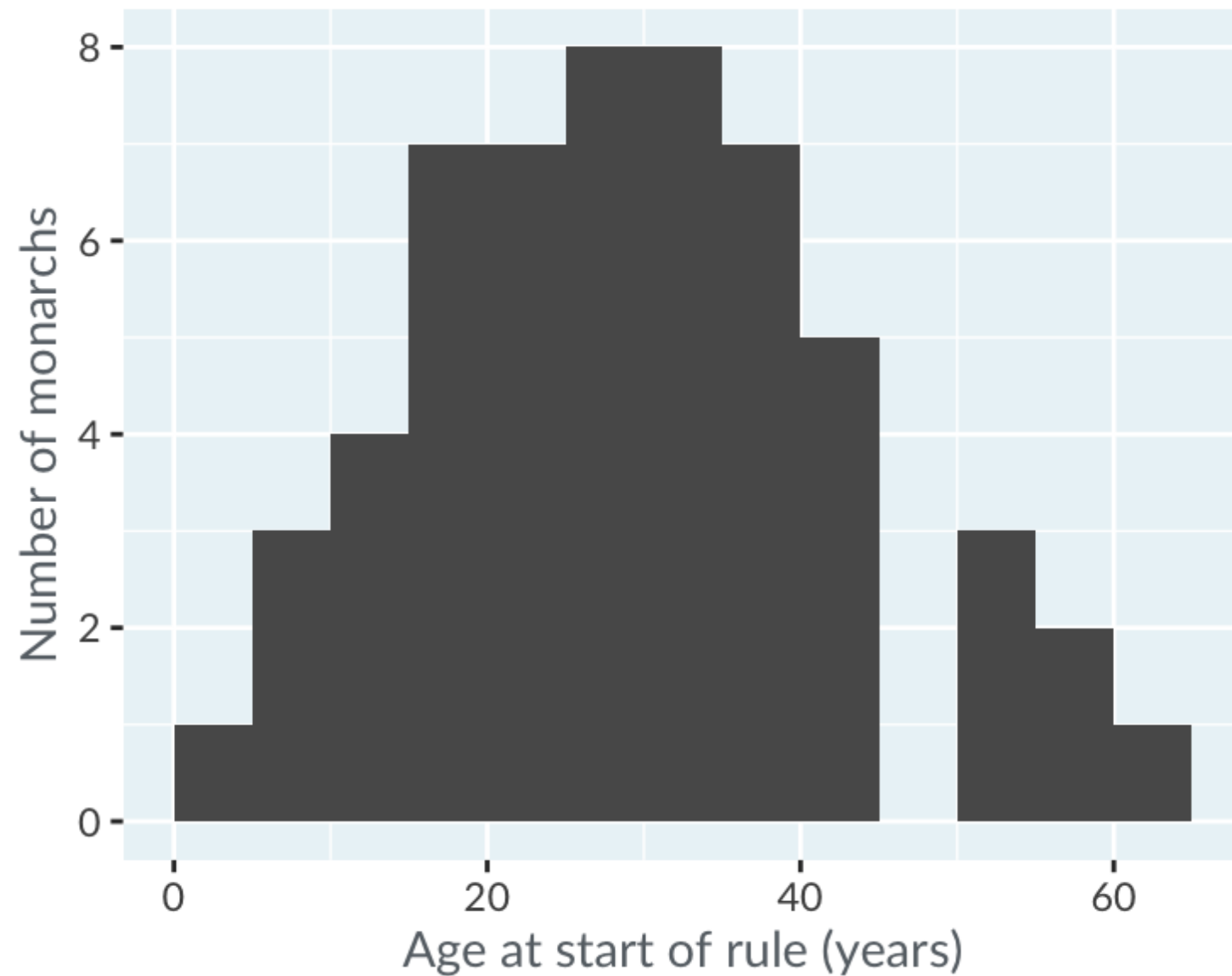


# Kurtosis: how many extreme values?





# Kurtosis: how many extreme values?





# Let's practice!

UNDERSTANDING DATA VISUALIZATION

# Box plots

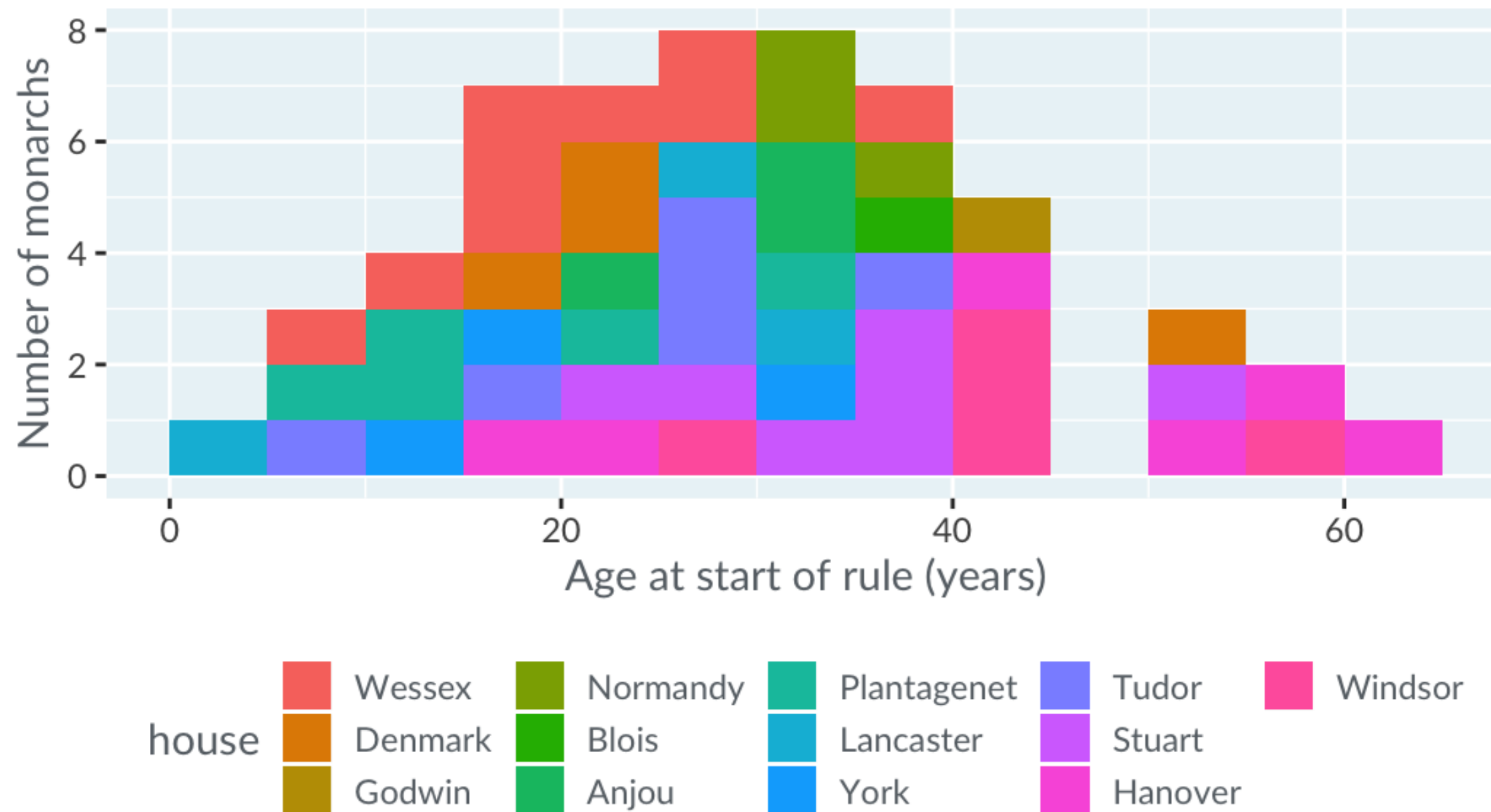
UNDERSTANDING DATA VISUALIZATION



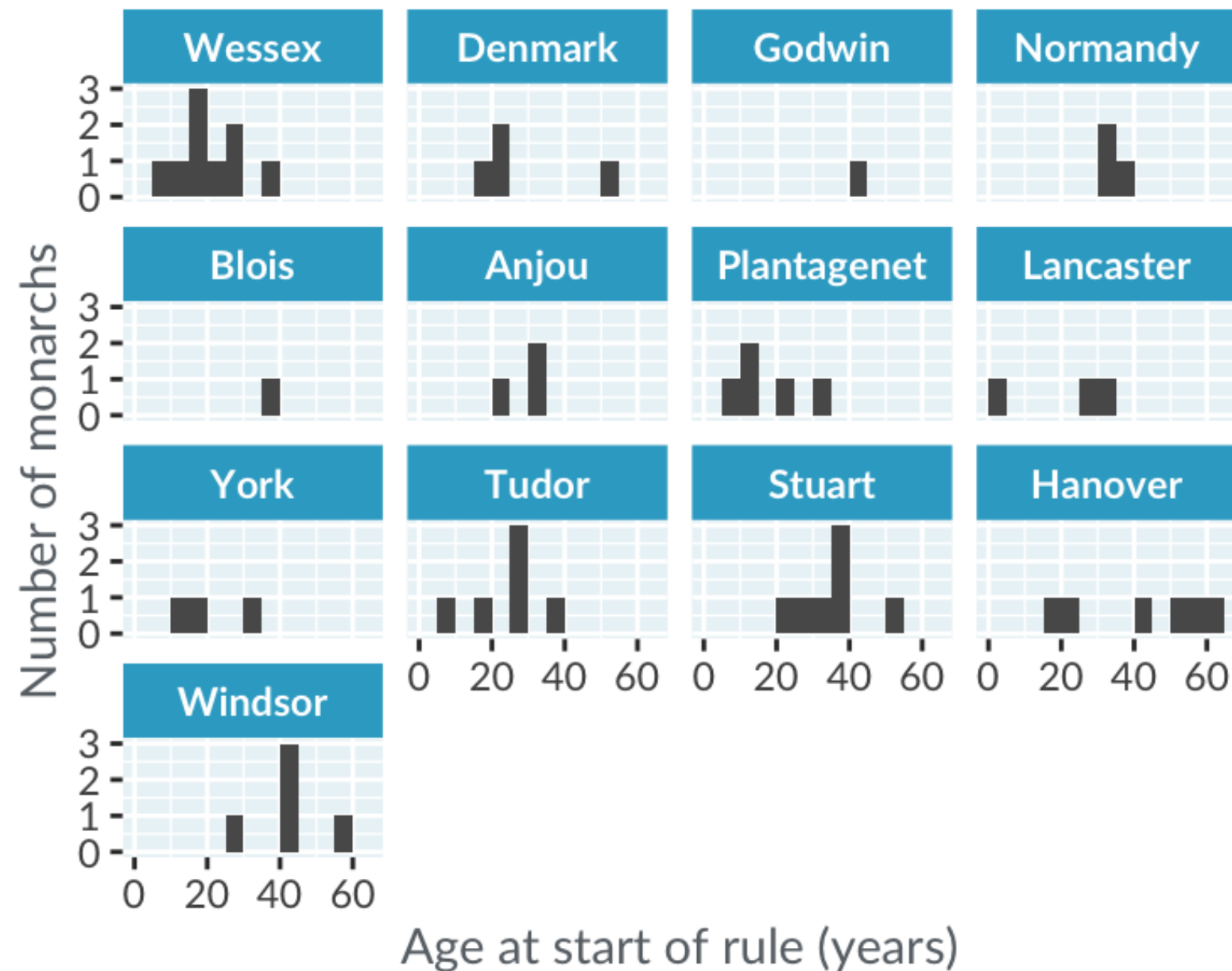
**Richie Cotton**

Data Evangelist at DataCamp

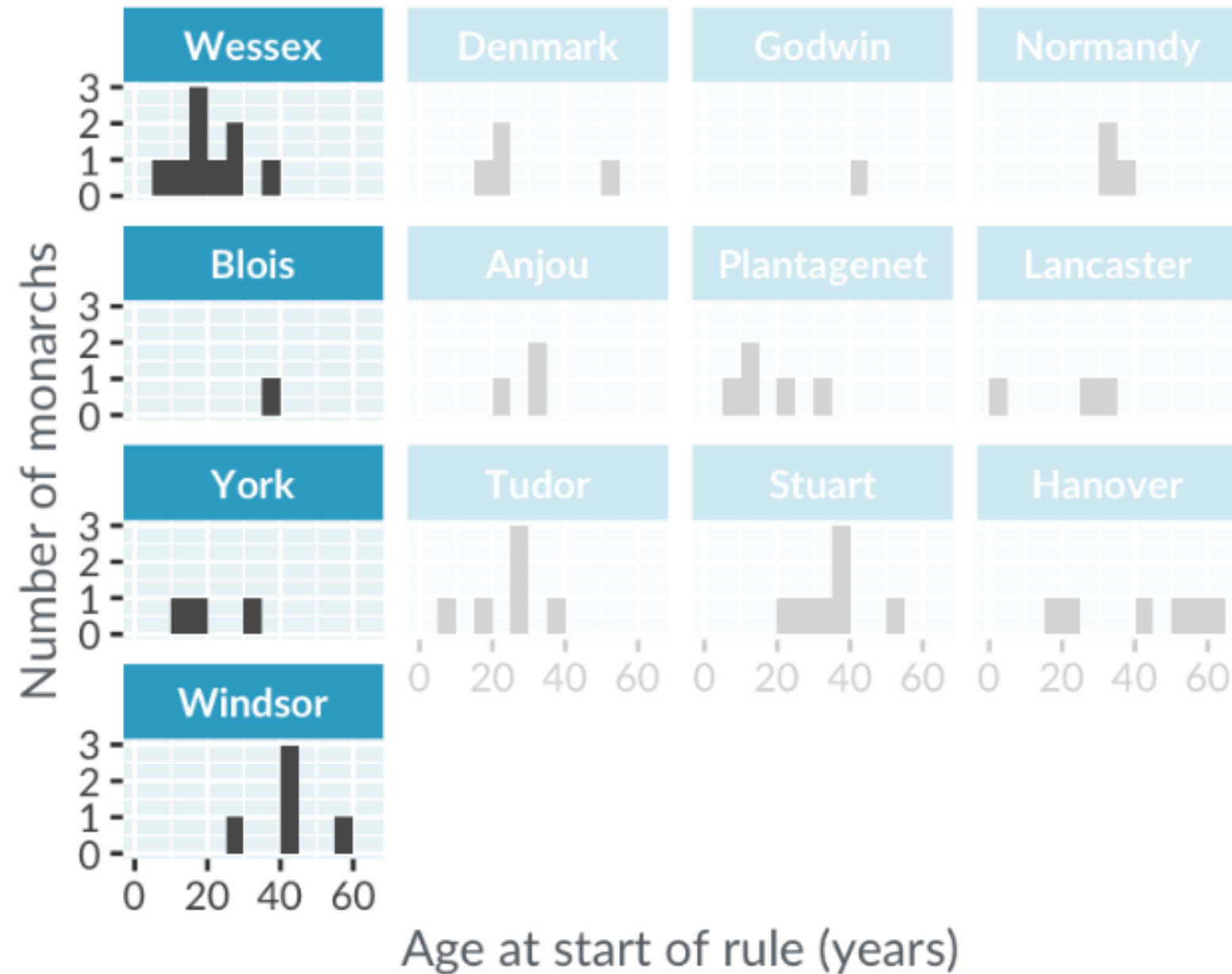
# You can't just color in histograms



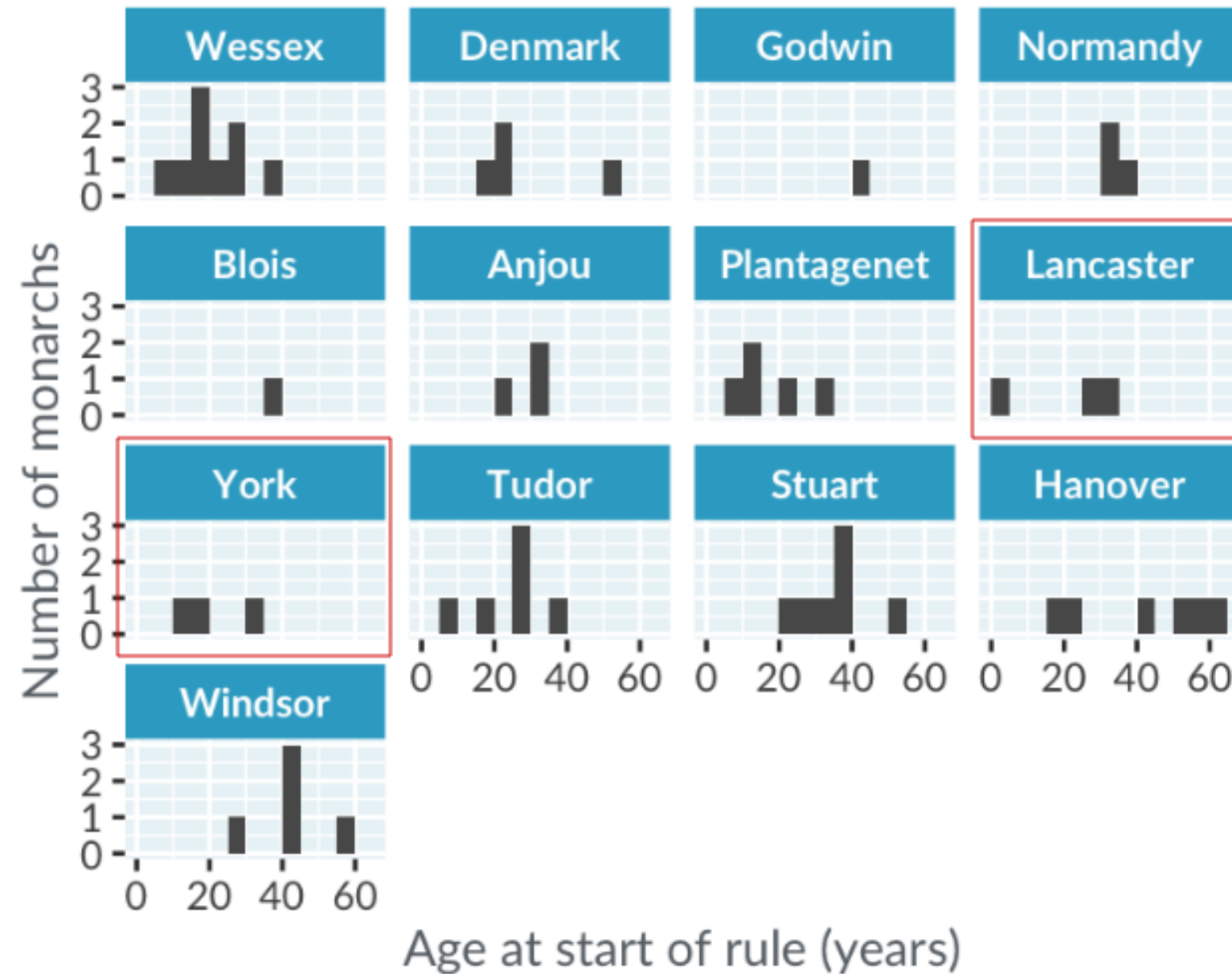
# Draw each histogram in its own panel

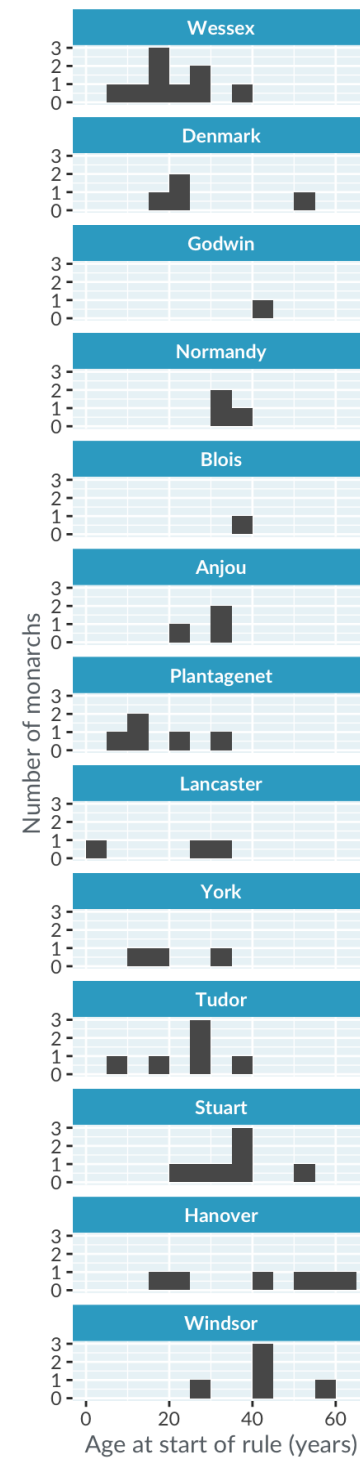


# Draw each histogram in its own panel



# Draw each histogram in its own panel





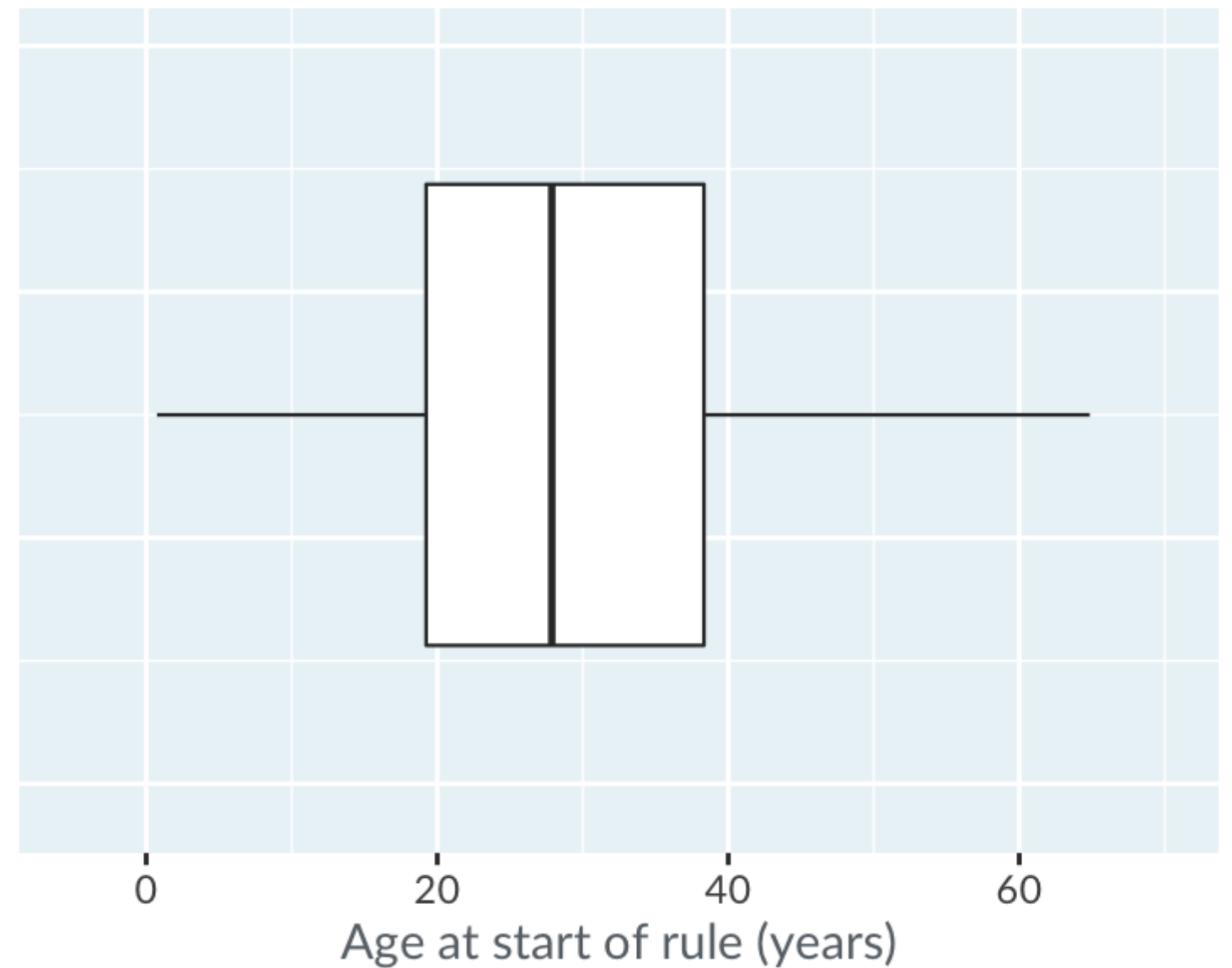
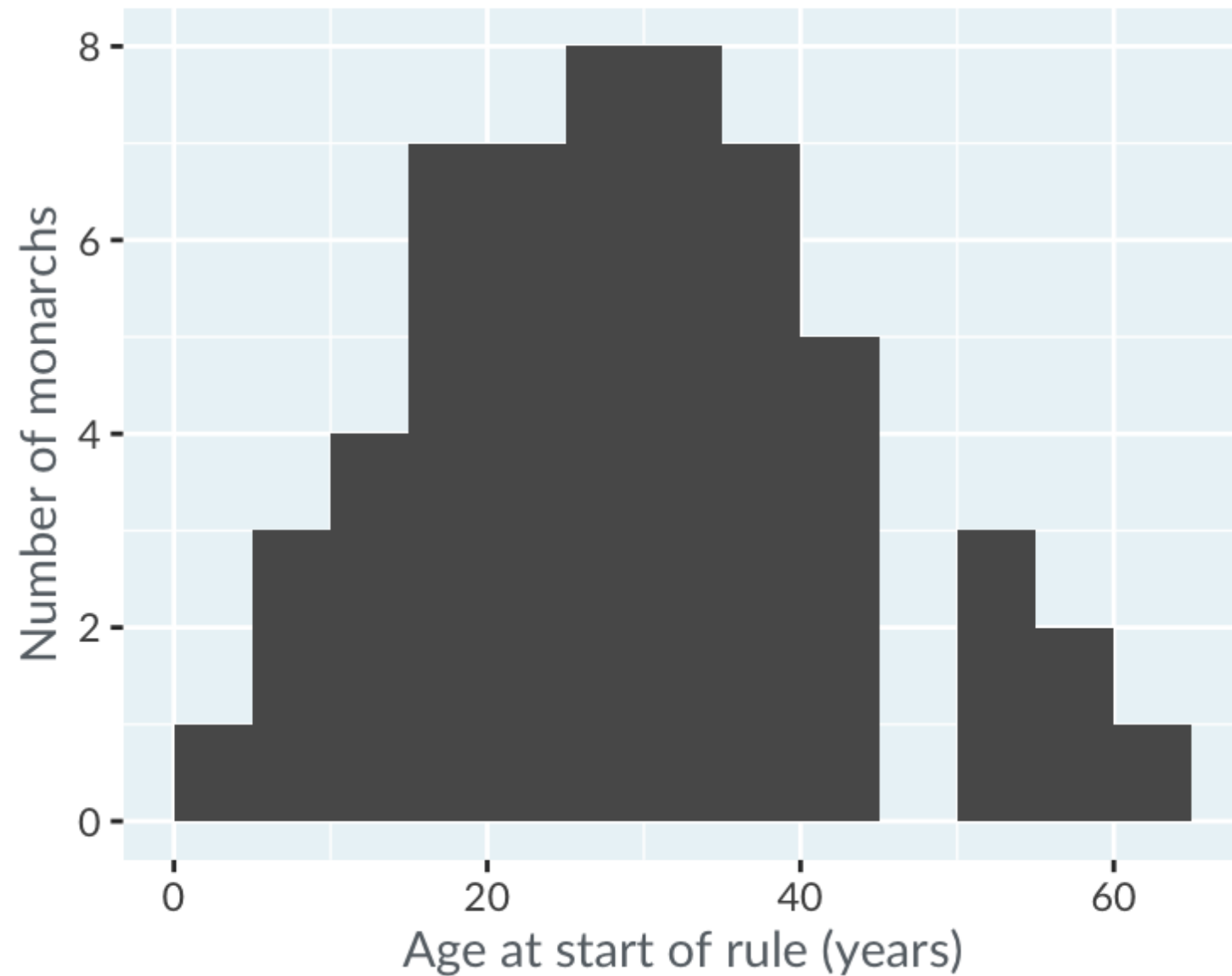


# When should you use a box plot?

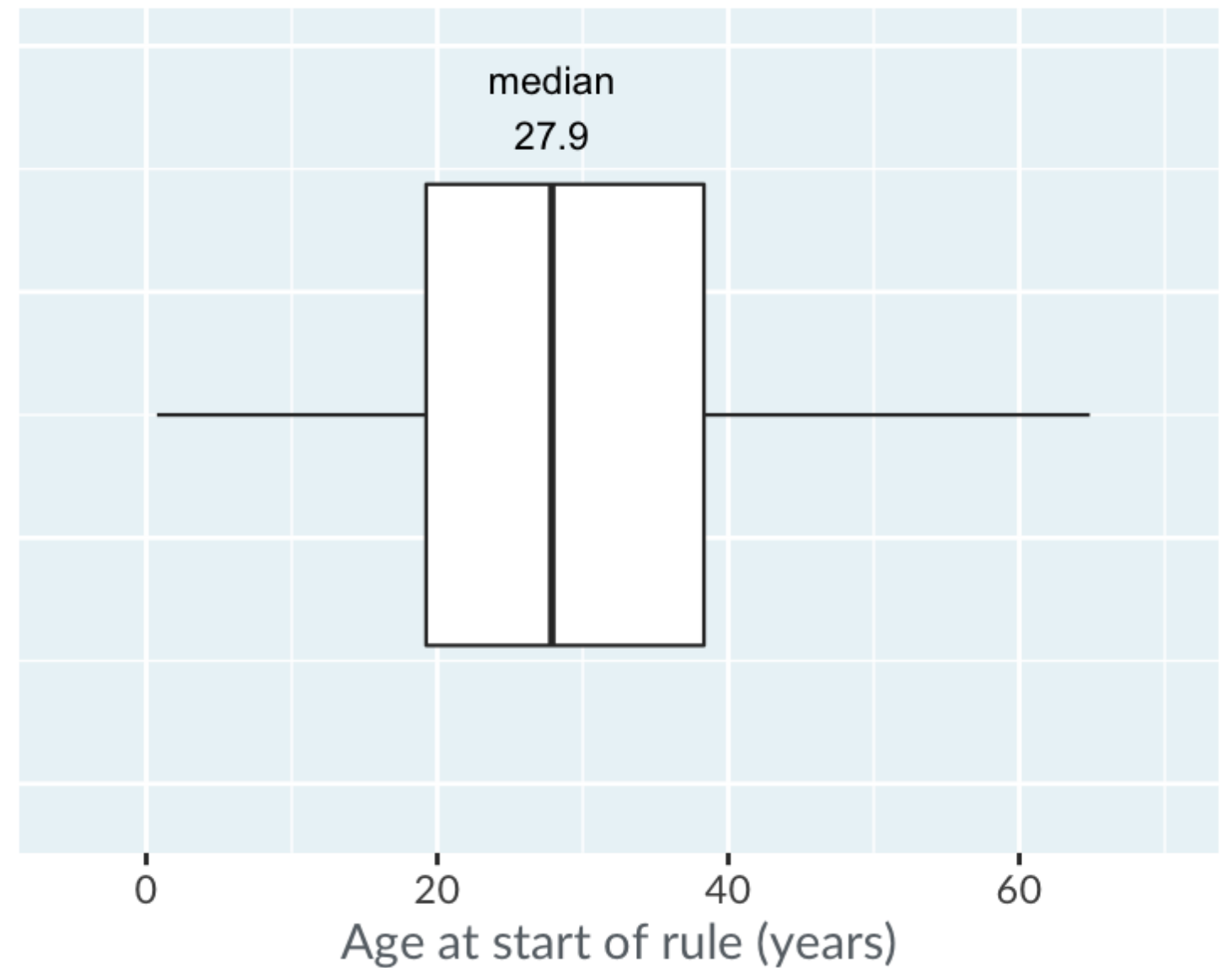
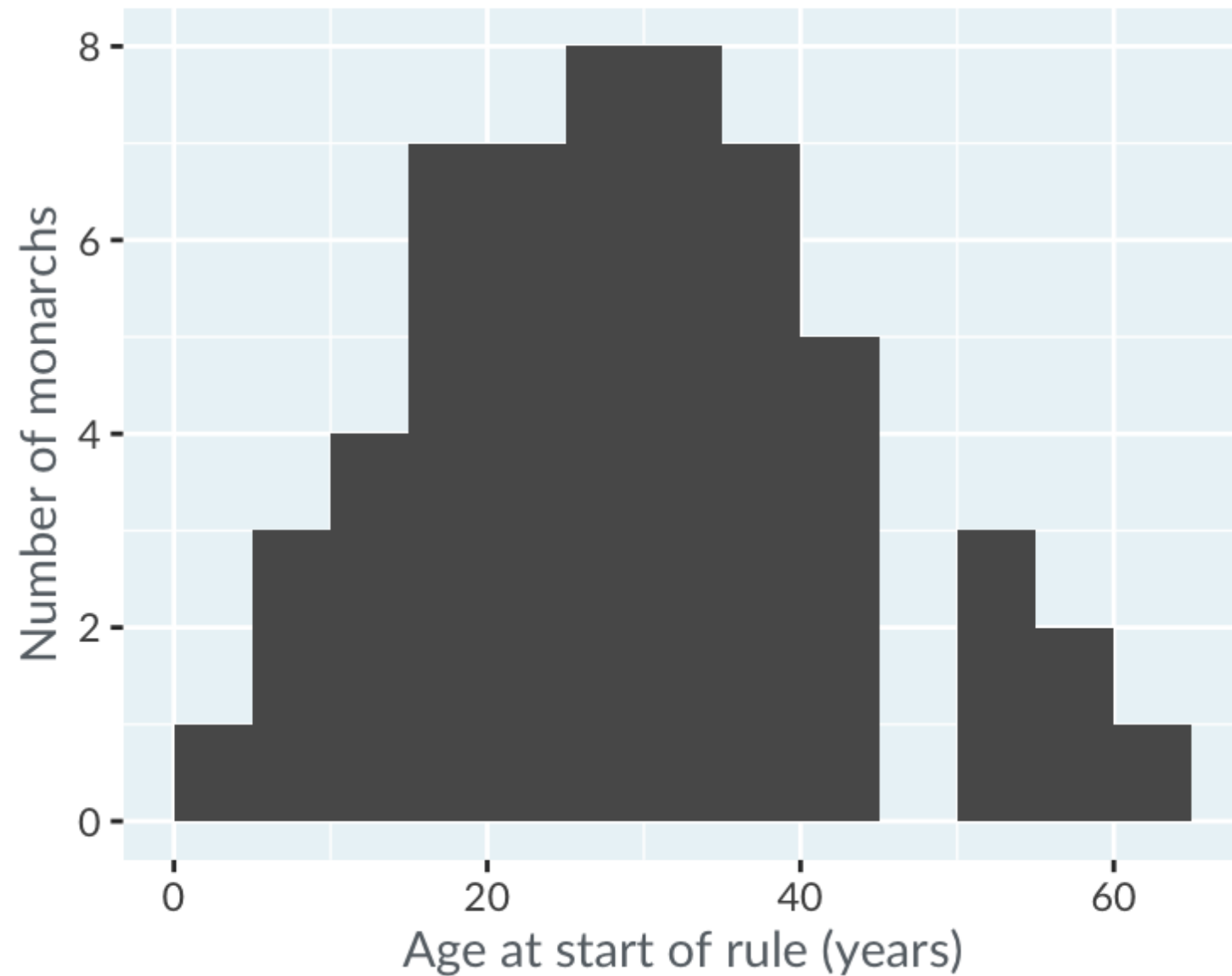
1. When you have a continuous variable, split by a categorical variable.
2. When you want to compare the distributions of the continuous variable for each category.



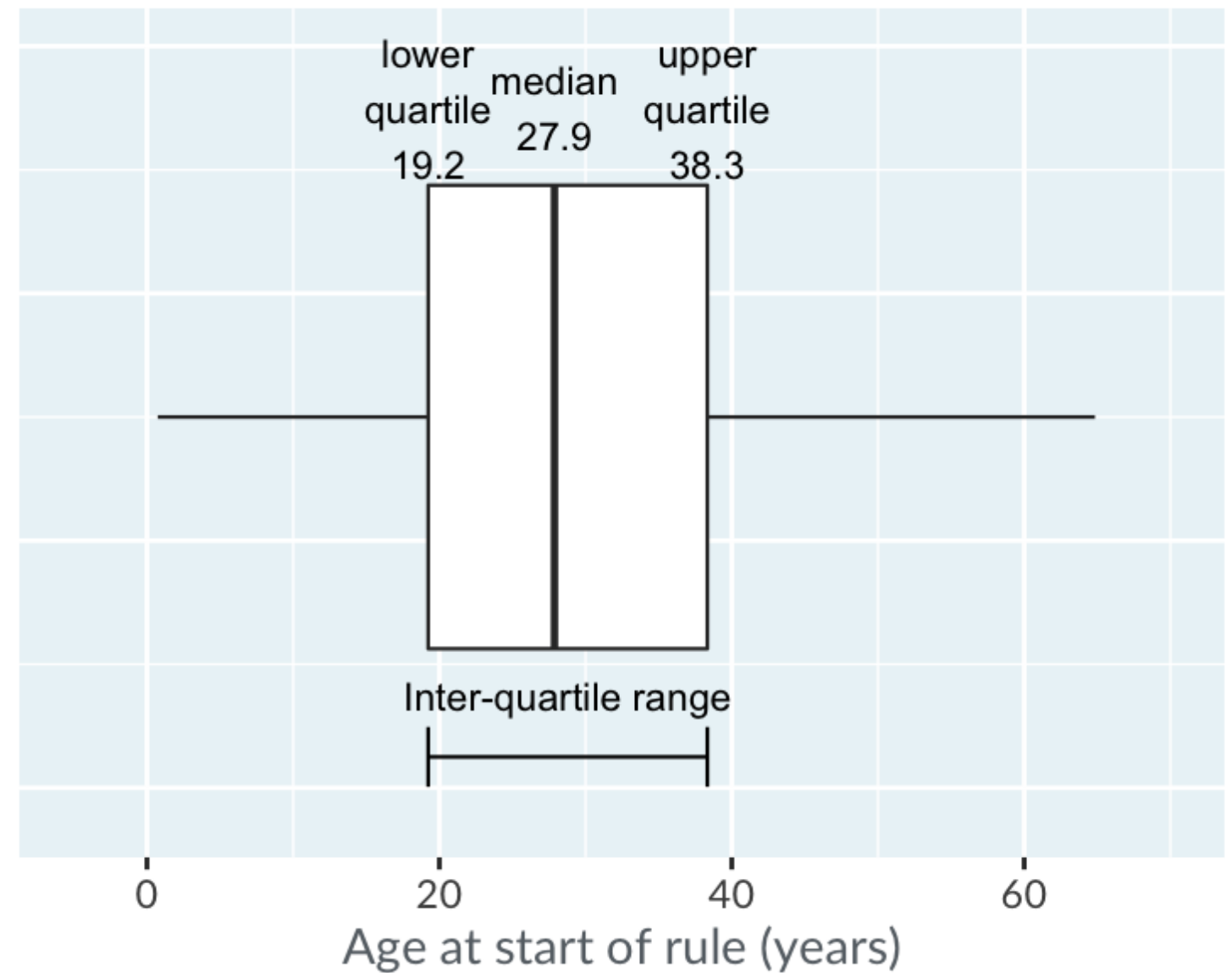
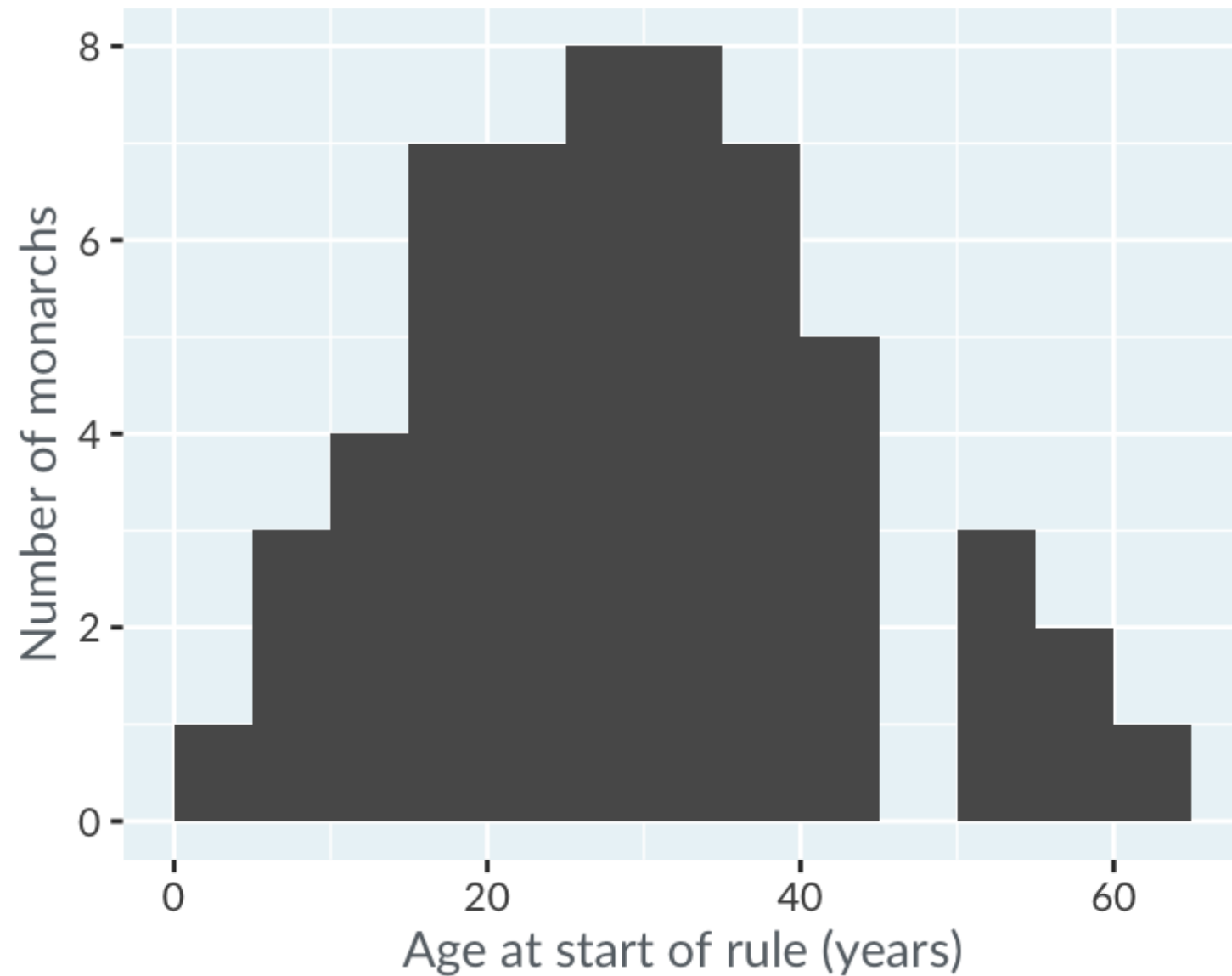
# Histogram vs. box plot



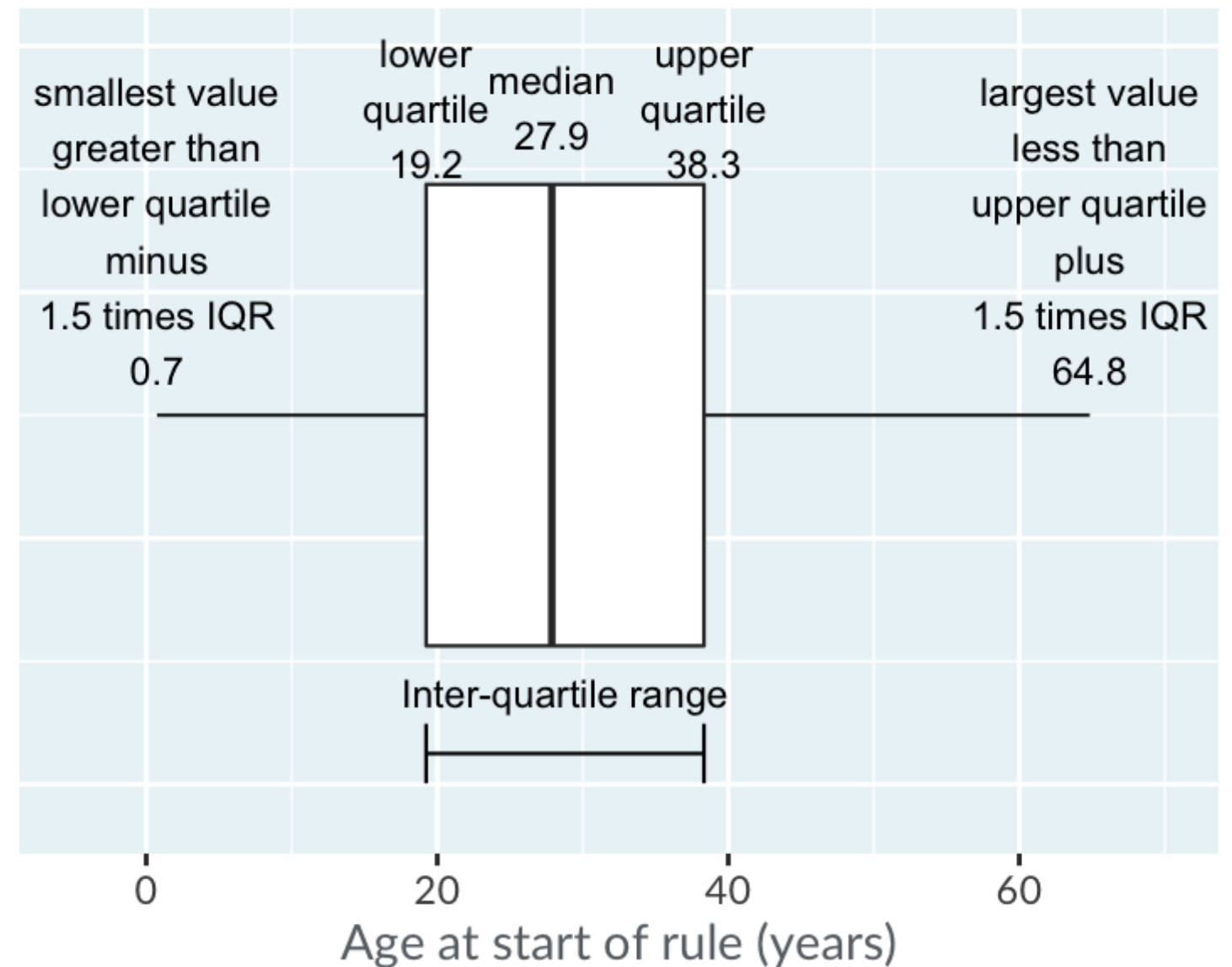
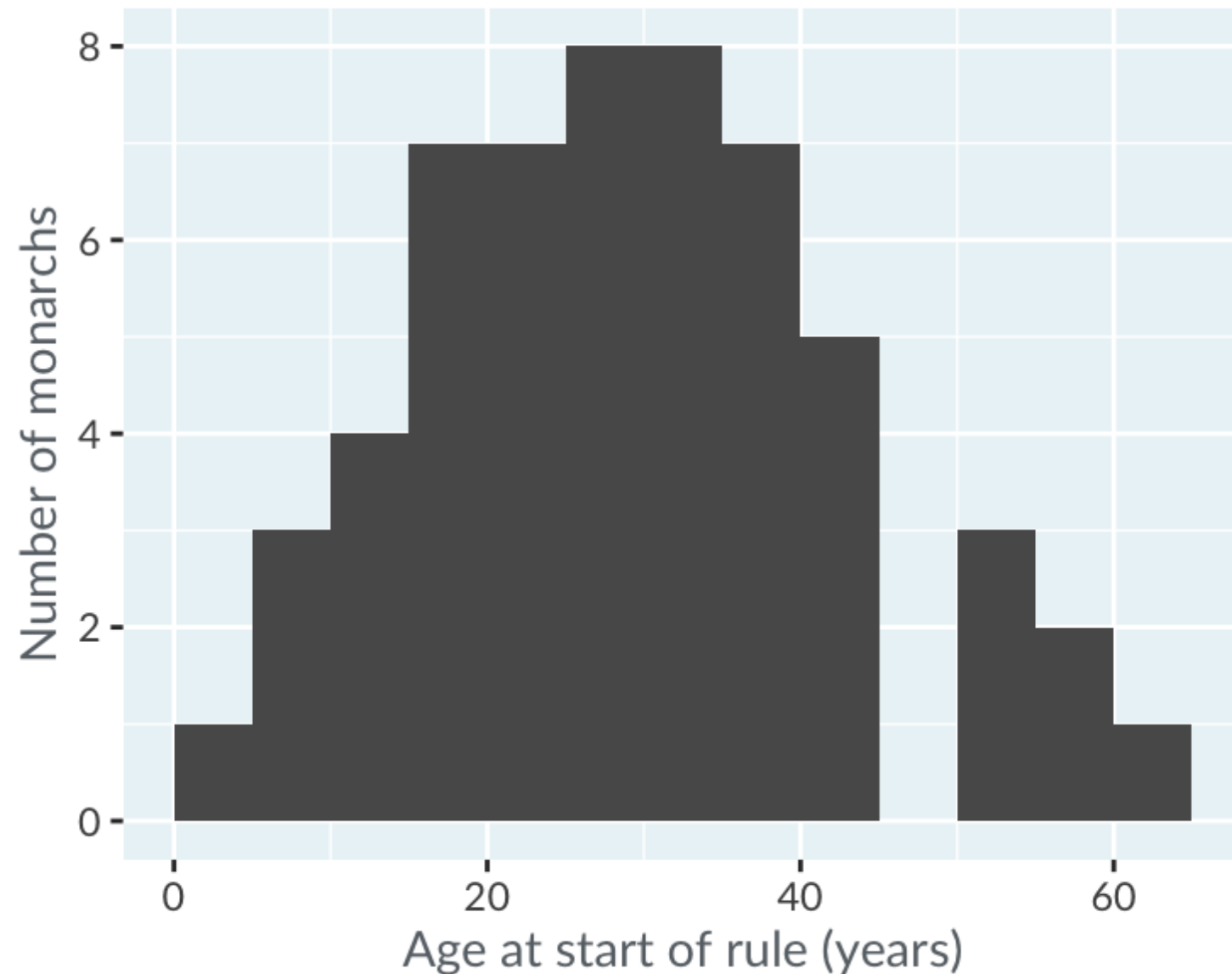
# Histogram vs. box plot: mid-line



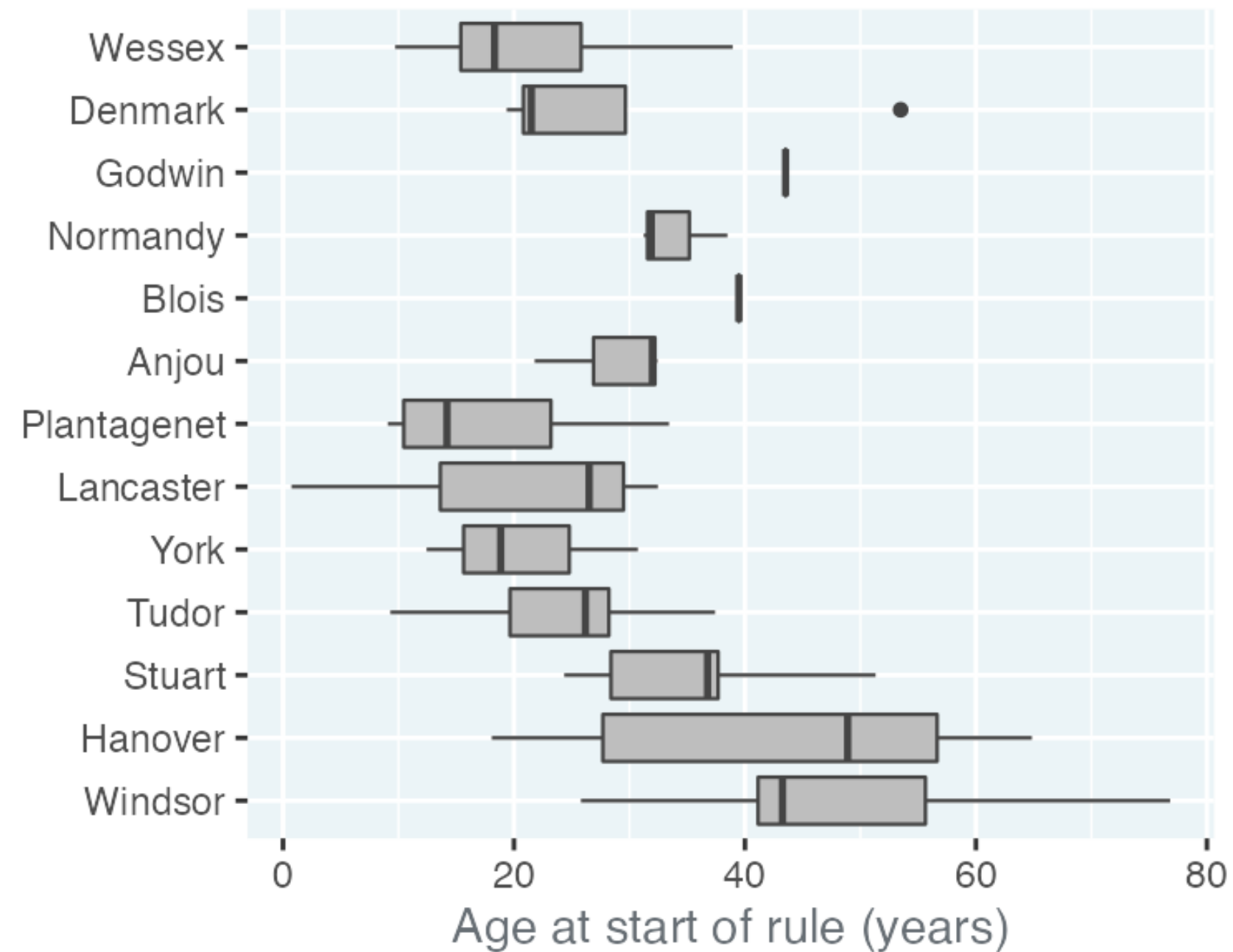
# Histograms vs. box plot: the box



# Histograms vs. box plots: the whiskers



# Monarchs by house



# Let's practice!

UNDERSTANDING DATA VISUALIZATION